ASDXDDL8M3 Data Modelling and Management Day 2 Summary

Tuesday, October 20th, 2026

Overview

Key participants include QA Trainer Ninad Rampure leading the session, with contributions and confirmations from learners such as Andrew, Ryan, Charlie, Helen, Sharon, Robin, Chris, Rachel, Ellen, Jonathan, and others. The meeting focuses on core relational database modeling concepts (many-to-many relationships and bridging tables), normalization (1NF through 3NF), data types and attribute definition, primary and foreign keys, ERD creation, and the importance of a data dictionary.

Major outcomes and decisions: the group agrees to use bridging (junction) tables for many-to-many relationships and to enforce normalization to at least 1NF before progressing to 3NF for import, as emphasized by Ninad. Primary keys and surrogate keys will be used where appropriate (noted by Andrew and reinforced by Ninad), and foreign keys will define relationships and cardinalities. The trainer commits to demonstrating two deployment approaches (pure SQL and a GUI-assisted flow) in the next session and to monitoring breakout groups while they populate a shared collab board.

Commitments and next steps: learners will copy their Word-based entity and attribute work into the shared whiteboard, complete ERDs, and finalize data types using the references Ninad posts in chat; Ninad will run day-two practical exercises focused on SQL DDL/DML (CREATE, INSERT, UPDATE, DELETE, ALTER, SELECT, joins, indexing) and demonstrate how ERDs and data manipulation connect to the apply-stage deliverables. The trainer also provides hands-on support for tool access and resolves breakout assignment issues during the session.

Concerns and notes: several participants experience temporary connection or navigation issues on the collab board, which Ninad addresses with refresh/rejoin guidance and reassignment where needed. Overall, the group aligns on modeling principles and is prepared to proceed into hands-on SQL deployment and collaborative ERD completion in breakout work.

Detailed Summary

many-to-many relationships

The group discusses identifying many-to-many relationships and using bridging (junction) tables to consolidate redundant data. This matters because many-to-many relationships create repeated data entries and require an intermediate table to

maintain correct relational structure. The current state is that learners recognize bridging tables are necessary and that automation of these tables is desirable.

- QA Trainer Ninad Rampure explains that many-to-many relationships require bridging tables to bring redundant data together.
- Charlie emphasizes that bridging tables and related data handling should be automated, according to QA Trainer Ninad Rampure.
- Robin and others confirm that bridging tables are the correct approach for many-to-many cases, as noted by QA Trainer Ninad Rampure.

normalization principles

The session reviews normalization, starting with First Normal Form (1NF) and progressing to 2NF and 3NF to remove duplication and improve data integrity. This is important because raw collected data arrives in inconsistent formats, and normalization standardizes structure to enable reliable database import and querying. Learners applied normalization concepts to example exercises like the movie database and plan to further decompose into third normal form.

- QA Trainer Ninad Rampure stresses the need to enforce at least First Normal Form when processing raw data.
- Ninad notes the movie database and the many-to-many exercise are already in 1NF and are being segregated toward 3NF for import.
- Ninad highlights that normalization reduces duplication and improves data integrity during modeling and deployment.

data types and attributes

The group covers attribute data types and how they mirror familiar concepts from spreadsheets (numeric, text, date, currency) and determine how the database stores values. This matters because setting proper data types prevents downstream processing issues and supports accurate querying and indexing. The current state is that learners received links and guidance to choose appropriate SQL data types for each attribute.

- Ryan and QA Trainer Ninad Rampure discuss specifying data types for attributes to match expected column formats.
- Ninad draws a parallel between Excel column types and database attribute types to clarify expectations to learners.
- Ninad shares data type reference links in chat so groups can finalize attribute definitions during breakout work.

primary and foreign keys

The discussion covers primary keys (natural and surrogate) and foreign keys as the mechanism to link tables and define cardinality. This matters because unique identifiers and foreign key relationships underpin referential integrity and determine

one-to-one, one-to-many, or many-to-many cardinalities. Learners are applying these ideas to their tables and considering surrogate keys where natural IDs are absent.

- Andrew explains that a primary key is a unique identifier for each record, illustrated by movie or actor IDs as mentioned by QA Trainer Ninad Rampure.
- Ninad notes surrogate keys may be introduced when natural identifiers do not exist, as learners experienced with movie IDs.
- Ninad clarifies that foreign keys indicate links between tables and help determine relationship cardinality such as one-to-many or many-to-many.

ERD and cardinality

Entity Relationship Diagrams (ERDs) and cardinalities are emphasized as central planning tools to visualize entities, keys, and their relationships. This matters because ERDs guide table design, attribute placement, and subsequent deployment; they also provide documentation for handover. Learners are expected to create ERDs from their identified entities and attributes as part of the modeling process.

- QA Trainer Ninad Rampure states that ERDs show entities, primary/foreign keys, and cardinalities and serve as the core planning artifact.
- Helen and Ellen emphasize that understanding entities and their links lets you start drawing ERDs even before finalizing attributes.
- Ninad recommends producing ERDs as part of documentation so future maintainers can pick up the database design.

data dictionary importance

The class discusses maintaining a data dictionary to document table attributes, data types, and ERD details so future team members can understand the schema. This matters because documentation preserves institutional knowledge and supports deployment and handover. The current state is that the trainer highlights the data dictionary as a required artifact for the modeling and deployment stages.

- QA Trainer Ninad Rampure explains that the data dictionary documents attributes and data types and enables others to take over database management.
- Sharon states that the data dictionary is important alongside ERD, keys, junction tables, and cardinalities.
- Ninad reiterates that the data dictionary will be needed during deployment and for ongoing maintenance or role changes.

deployment and SQL practice

Day two focuses on hands-on deployment using pure SQL queries to create databases, tables, insert and manipulate data (CREATE, INSERT, UPDATE, ALTER,

DELETE), plus querying and indexing. This matters because practical SQL application demonstrates the full lifecycle from model to implementation. Learners will practice both SQL-first and GUI-assisted deployment flows and are not required to finish exercises within the session timeframe.

- QA Trainer Ninad Rampure outlines that day two covers creating databases/tables and using SQL DDL/DML statements for deployment.
- Ninad says he will demonstrate two deployment approaches: SQL queries and a graphical interface, and that ERD can be created before or after data insertion.
- Ninad clarifies there is no strict submission expectation and learners can practice SQL at their own pace to cover apply-stage requirements.

breakout activity using whiteboard

Learners move into breakout groups to finish the word-based activity: list five entities, define attributes and data types, and assemble ERDs on a shared online whiteboard (collab board). This matters because collaborative editing and shared visibility accelerate review and reduce barriers for those unfamiliar with screen sharing. The current state is groups are instructed to copy/paste content into the whiteboard, use navigation (right-click to pan), and edit tables collaboratively while trainer monitors.

- QA Trainer Ninad Rampure instructs groups to copy-paste their Word content into the shared collab board rather than retyping.
- Ninad demonstrates navigation controls (right-click to pan, click once to select, twice to edit) and row/format editing on the whiteboard.
- Ninad assigns groups and starts a 30-minute breakout window for completing diagrams and table contents, while monitoring and reassigning participants as needed.

tool access and troubleshooting

The trainer shares links to the collab board and data type references and helps participants troubleshoot connectivity and navigation issues. This matters because seamless access ensures productive group work during breakouts. The current state is that most participants connect successfully after refreshing or rejoining, while the trainer monitors and assists individuals to join correct group rows.

- QA Trainer Ninad Rampure shares access links in chat and explains guests can join without signing up to the portal.
- Chris reports a websocket connection message and resolves it by refreshing per Ninad's suggestion.
- Ninad actively helps participants refresh, reassign breakout membership, and demonstrates how to move to the correct row on the whiteboard.